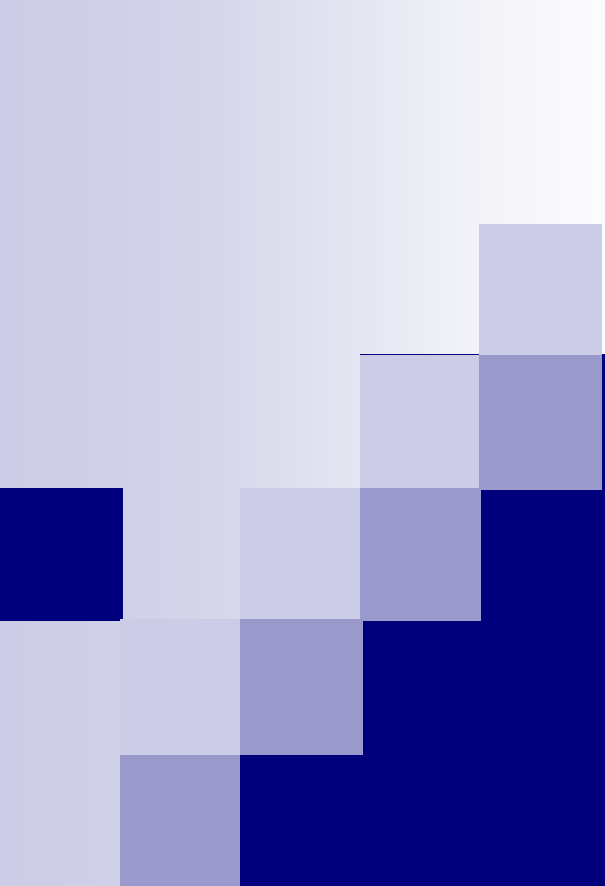




Modern Operating Systems

Chapter 0 - Preface

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Reference Books (1)

■ Required Textbook

- *Modern Operating Systems*, 4th edition, Andrew S. Tanenbaum, 机械工业出版社, 2017

■ Reference Books

- *操作系统导论, Operating Systems: Three Easy Pieces*, Remzi H. Arpaci-Dusseau & Andrea C. Arpaci-Dusseau, 人民邮电出版社, 2019
- *操作系统：原理与实现*, 第2版, 陈海波&夏虞斌, 机械工业出版社, 2023



Reference Books (2)

■ Reference Books (ctd.)

□ *Operating Systems: Design and Implementation, 3rd Edition*, 电子工业出版社, 2007



□ *openEuler 操作系统, 第2版*, 任炬&张尧学, 清华大学出版社, 2022





Course Prerequisites

- Hardware

- Computer Organization and Architecture

- Software

- Data Structure

- C/C++

Why are Operating Systems Important? (1)

- Important to understand and know how to correctly use when writing user applications.
 - Kernel functions
 - Concurrent programming, e.g thread shared memory and thread lock
 - Multithread
 - Lock
 - Deadlock
- Large and complex systems that have a high economic impact and result in interesting problems of management.
 - Resource management
 - Performance and conflict

Why are Operating Systems Important? (2)

- Few actually involved in OS design and implementation but nevertheless many general techniques to be learned and applied.
 - Database, Cloud system
 - Also need Memory management, File system
- Learn how to design a software system.
 - Design Tradeoffs
 - Reliability
 - Performance
 - Maintainability
 - Portability
 - ...



Course Outline

- Chapter 1 Introduction
- Chapter 2 Processes and Threads
- Chapter 6 Deadlocks
- Chapter 3 Memory Management
- Chapter 4 File Systems
- Chapter 5 Input/Output

课程特点

- 实践性强（从实践总结出原理）
- 涉及面广（并行程序，性能问题，结构问题，程序方法论，软件工程，等等）
- 如何学好操作系统？
 - Learn OS concepts by coding them!
 - Linux 0.11 kernel code
 - MINIX source code
 - 清华大学ucore
 - openEuler社区

学习要求

- 按时上课，认真听讲
- 积极主动阅读参考书
- 整理笔记
- 动手编码
- 思考，讨论，提问
- 及时沟通，反馈



Grading

- Final exam: 60%
- Homework: 20%
- 4 Labs : 20%